MILITARY SPECIFICATION

CLOTH, SPUN VISCOSE RAYON, RESIN IMPREGNATED

1. SCOPE

1.1 Scope. This specification covers three classes of cloth designated as Cloth, Spun Viscose Rayon, Resin Impregnated, for use with ammunition.

1.2 Classification. The cloth shall be of the following classes as specified (see 6.2):

Class 1—lighter weight
Class 2—heavier weight
Class 3—scarlet colored

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-270—Paper, Wrapping, Dry Waxed
CCC-T-191—Textiles; Test Methods
PPP-B-601—Boxes, Wood, Cleated Plywood
DDD-S-751—Stitches, Seams and Stitching
PPP-B-585—Boxes, Wood Wirebound (Overseas Type)
PPP-B-621—Boxes, Wood, Nailed and Locker Corner

MILITARY

MIL-L-10547—Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible

STANDARDS

FEDERAL

FED-STD-595—Colors

MILITARY

MIL-STD-105—Sampling Procedures and Table for Inspection by Attributes
MIL-STD-109—Inspection Terms and Definitions
MIL-STD-129—Marking for Shipment and Storage
MIL-STD-1235—Single and Multilevel Continuous Sampling Procedure and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification 21—Rating, Rules and Regulations Association of American Railroads

(Applications for copies of these freight classification rules should be addressed to the Secretary of Operating Transportation Division, 59 Van Buren Street, Chicago, Ill.)

NATIONAL MOTOR FREIGHT CLASSIFICATION
No. A-2—Naming Uniform Rating Rules and Regulations

(Copies may be obtained from the Issuing Officer, 1424 16th Street, N. W., Washington 6, D. C.)

FSC 8305

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3. REQUIREMENTS

3.1 Material. The cloth shall be woven from spun viscose rayon yarn.

3.2 Finish (see 6.3). The cloth shall be finished with a urea formaldehyde resin (past type) plus a durable water repellant (melamine resin base) when tested as specified in 4.3.2.

3.3 Width. The width of the cloth shall be as specified in the contract or order, when determined as specified in table II.

3.4 Weave. The weave shall be a plain, one end in dent, single weave, when determined as specified in 4.2.2.1.

3.5 Physical characteristics. The finished cloth shall comply with the limits specified for the characteristics listed in Table I, when tested as specified in 4.3.1 and table II.

TABLE I

	Class 1		Class 2 and 3	
	Minimum (Min)	Maximum (Max)	Min.	Max
Weight, ounces per square yard	2.80	3 20	6 75	7 25
Breaking strength, pounds per inch.				
Warp	35		35	
Filling	35		85	
Thread count				
Warp	48		35	
Filling	48		35	

3.6 Breaking strength loss. The loss in breaking strength of the resin inpregnated fabric conditioned in dinitrogen tetroxide shall be not more than 10 percent, when tested as specified in 4.3.3.

3.7 Slippage value.

3.7.1 *Warp.* The slippage value of the warp yarns shall be a min., of 60 when tested as specified in 4.3.4.

3.7.2 *Filling.* The slippage value of the fill yarns shall be a min. of 60, when tested as specified in 4.3.4.

3.8 Stretch.

3.8.1 *Warp.* The stretch of the warp yarns shall be a max. of 10.0 percent, when tested as specified in 4.3.5.

3.8.2 *Filling.* The stretch of the fill yarns shall be a max. of 10.0 percent, when tested as specified in 4.3.5.

3.9 Ash content. The ash content shall be 1.0 percent max., when tested as specified in 4.3.6.

3.10 pH of water extract. The pH of water extract shall be 5.5 min. to 7.5 max., when tested as specified in table II.

3.11 Standard spray test rating. The standard spray test rating of the cloth shall be 70 min., when determined as specified in table II.

3.12 Color.

3.12.1 *Class 1 or 2.* Unless otherwise specified in purchase order or contract the color shall be natural. When a color is specified, the dye or tint shall be fugitive (see 6.5) when tested as specified in 4.3.7.

3.12.2 *Class 3.* The cloth shall be scarlet.

3.13 Workmanship. The finished cloth shall be clean, evenly coated and shall conform to the quality and grade of product established by this specification. The maximum and minimum diameter of the roll shall be specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible

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for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 *Contractor quality assurance system.* If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring and testing equipment) and provisions for control and disposition of nonconforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractors system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
(b) Quantity of product inspection.
(c) Results obtained for all inspection performed.
(d) Specification number and date, together with an identification and date of changes.
(e) Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
(f) Quantity of product in the lot.
(g) Date submitted.

The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agents authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractors written quality assurance procedure (see 4.1.1.), this detail specification, and other contractual documents as a guide, the Government inspector shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with (a) or (b), as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that

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all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector and, depending upon the nature of the deviation, may consist of lot rejection, screening, resampling, re-inspection of the suppliers employees, or other appropriate action.

(a) Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.3.2.

(b) Verification at the point of delivery shall be in accordance with 4.1.3.2.

4.1.3.1 *Surveillance.* Surveillance shall include, but is not limited to:

(a) Observation of procedures concerning lot formation and identification.

(b) Observation of sampling procedures and application of acceptance criteria.

(c) Determination that all required examinations and tests are performed in accordance with prescribed procedures of this specification, or approved equivalents thereto.

(d) Review of procedures for control and disposition of nonconforming material.

4.1.3.2 *Product inspection.* Product inspection shall consist of Government inspection of product which has been inspected by the contractor and found to meet the quality assurance provisions of this specification to determine that the product is of the quality required by this specification and that the contractors records are reliable.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of cloth of one class only, produced by one manufacturer, under one continuous set of operating conditions, in accordance with the same specification or same specification revision.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect.

4.2.2.1 *Cloth in bolts prior to packing.*

Categories	*Defects*	*Method of inspection*	*Code No (see 6.7)*
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Weave improper	Visual	01001
102.	Holes, tears, dropped or loose threads	Visual	01002
103.	Blisters or lumps	Visual	01003
104.	Uneven coating (thin areas where coating compound is missing or insufficient)	Visual	01004
Minor:	AQL 1.00 percent		
201.	Stain or streaks (coating compound stain or streak clearly noticeable	Visual	01005
202.	Uneven coating (heavily coated areas, clearly noticeable)	Visual	01006

4.2.2.2 *Open box.*

Categories	*Defects*	*Method of inspection*	*Code No*
Critical: None defined.			
Major:	AQL 0.25 percent		
101.	Bolts not individually wrapped	Visual	02001
102.	Box liner material improper	Visual	02002
Minor:	AQL 1.00 percent		
201.	Liner not sealed	Visual	02003
202.	Liner torn or pierced	Visual	02004

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4.2.2.3 *Closed box.*.

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Gross weight, max.	Approved scale	03001
102.	Marking misleading or unidentifiable	Visual	03002
103.	Boxes improperly closed	Visual	03003
Minor: None defined.			

4.2.3 *Tests.*

4.2.3.1 *Sampling.*

4.2.3.1.1 *For all classes.* The sample size shall be in accordance with Standard MIL-STD-105, Level I.7 of table III-B. From each roll or piece 2 yards in length shall be selected. If any sample fails to comply with any requirement, the lot shall be rejected.

4.2.3.1.2 *Dyed cloth.* A sample consisting of five pieces each one foot in length, shall be selected from each lot for the finish test.

4.3 **Test methods and procedures.** The following test shall be performed in accordance with referenced method or as specified herein.

4.3.1 *Testing in accordance with Specification CCC-T-191.* The test specified in table II shall be conducted in accordance with the specified method contained in Specification CCC-T-191.

TABLE II

Test	Specification CCC-T-191 Code	
	Method	No.
pH of Water extract	2811	01001
Width	5020	05001
Weight	5040	06001
Thread count	5050	07001
Breaking strength	5100	08001
Spray test	5526	09001

4.3.2 *Finish—Code No. 10001.* A finish sample of undyed cloth approximately 6 inches, shall be dyed in a bath containing disperse dye (Eastman Brilliant Pink 3B-GLF[1] or equal) and a direct dye (Chlorantine Fast Green 5GLL[1] or equal) in the ratio of 2:1. The dyeing conditions are as follows:

8 grams per liter Eastman Brilliant Pink 3B-GLF

4 grams per liter Chlorantine Fast Green 5GLL

Dyeing time: 3 minutes.

Dyeing temperature: 68° ± 2 degrees Centigrate (°C.).

If the color of the sample after dyeing is purple, the finish is satisfactory. If the color is green, the finish of the sample is not satisfactory. (see 6.8).

4.3.3 Determination of breaking strength loss (see 6.6)—Code No. 11001. Air is drawn through the apparatus shown in figure 1 in the direction indicated on the diagram, by an aspirator. A series of two drying towers, containing magnesium perchlorate with a small amount of indicating Drierite, serves as the air intake. In the diagram, the two drying towers are shown with separate outlets to the humidifier and the mixing bulb on the left. In practice, the towers are arranged in series with a glass "T" connecting them to the humidifier and mixing bulb. Dry air passes from one arm of the "T" tube attached to the second tower, to the humidifier containing a saturated salt solution selected to give the desired relative humidity. Dry air from the other arm of the "T" is admitted to a mixing bulb where it is mixed with dinitrogen tetroxide from the cylinder, shown in the upper left on the diagram. A portion of the resulting mixture is admitted to a second mixing bulb where it is mixed with air coming from the humidifier. Two needle valves control the flow of gases into each of the mixing bulbs; two flowmeters indicate the flow. The excess mixture from the first mixing bulb passes through two towers packed with pellets or flasks of solid sodium hydroxide to remove the dinitrogen tetroxide, and on out the aspirator. The gas from the second mixing bulb, which consists of air whose temperature is determined by the temperature of the room or enclosure in which

[1] Products of Eastman Chemical Products Inc., Kingsport, Tennessee

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the equipment is placed, and which has the desired dinitrogen tetroxide content, and relative humidity, flows over the specimens suspended in the exposure chamber. It then passes through the sodium hydroxide towers, and on to the aspirator. The pressure in the system is governed by the adjustment of the aspirator, and by the head of the ethylene glycol above the lower end of the open tube in the regulator in the lower right of the diagram. The exposure chamber is a 13-liter pyrex jar equipped with a ground glass cover. The specimens are supported by a stainless steel rack which is rotated at a rate of approximately 1 revolution per minute by a gear train and motor mounted on the cover. A wooden support holds the cover, and attached specimen rack when not in position on the exposure chamber. A plain glass cover is used to close the chamber when sweeping out the exposure train with air before, and after exposures. The two sampling parts shown in figure 1 provide for the removal of samples of air for chemical analysis from the exposure chamber, and from the stream of air entering the chamber. The humidifier consists of a plexiglass box 2 inches deep, 9 inches wide, and 12 inches long, filled to a depth of approximately 1 inch with a saturated salt solution. The box is equipped with baffles so that the air passing through it flows over a layer of solution one inch wide, and 9 feet long. A stoppered opening in the top of the box near the outlet provides for the insertion of a sensitive element of an electric hygrometer, if desired. The four flowmeters are of the ball-float type and, as shown in the diagram from left to right, have maixmum capacities of 42, 410, 116 and 960 milliliters per minute of air 70 degrees Fahrenheit at one atmosphere. The float of the meter used to measure dry air is of stainless steel; those of the other three meters are of the sapphire.

(a) Tensile testing machine. The tensile testing machine shall be as specified in Specification CCC-T-191, Method 5100.

(b) Apparatus for conditioning. Provisions for conditioning the specimens, and measuring their breaking strengths in air at specified temperature and relative humidity shall be in accordance with specification CCC-T-191, Textile Test Methods, Part 1, Section 4—Atmospheric Conditions for Testing.

(c) Cylinder of dinitrogen tetroxide (98 percent pure).

(d) Saturated magnesium nitrate solution for humidifier.

(e) Sodium hydroxide flakes or pellets for neutralizer.

(f) Ethylene glycol for pressure regulator.

(g) Magnesium perchlorate and indicating Drierite for drying tower.

4.3.3.1 *Preparation of specimen.* The specimens shall be prepared from each sample as follows: Cut two strips of cloth 28 inches long and 1½ inches wide at random from the fabric parallel to the wrap yarns. Cut two similar strips parallel to the filling yarns. Ravel each strip to a width of one inch by taking approximately the same number of yarns from each side. Mark the strips into four 7-inch specimens, labeling each specimen for identification. Cut the two end specimens from each strip and retain for controls. The 14 inch strips are to be used for exposure to dinitrogen tetroxide and are not to be cut into 7 inch strips until after exposure.

4.3.3.2 *Procedure.* The specimens (see 4.3.3.1) shall be tested as follows: Condition specimens at 21°C. and 65 percent relative humidity for at least 48 hours prior to exposure. Only two 14 inch strips (1 warp and 1 filling) of any one fabric are to be exposed to dinitrogen tetroxide at the same time. Strips of different fabrics may be exposed together (see 6.4). With the plain cover on the exposure chamber, and the needle valve between the dinitrogen tetroxide cylinder and flowmeter 1, shown on the left in figure 1, shown on the left in figure 1, closed, turn on the aspirator. Adjust the needle valves at flowmeters 2, 3 and 4 to give a stream of air having a relative humidity of 45 ± 3 percent at 27° ± 2°C., and rate of flow

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sufficient to change the atmosphere in the chamber four times an hour. Suspend the strips of fabric from one end of the specimen rack. After the aspirator has been running thirty minutes, replace the plain cover on the exposure chamber with the ground glass cover with attached rack, and strips of fabric. Start the specimen rack rotating. One hour after the introduction of the specimens into the exposure chamber, open the needle valve on flowmeter 1 halfway. Slowly open the outlet valve on the dinitrogen tetroxide cylinder until the reading on flowmeter 1 is approximately that required to give concentration of 0.2 percent dinitrogen tetroxide by volume to the stream of air entering the exposure chamber. Make the fine adjustment at flowmeter 1 with the needle valve. Make any necessary adjustments of the needle valves at flowmeters 2, 3 and 4 to maintain the relative humidity, and rate of flow. At the end of 2 hours from the time the flow of dinitrogen tetroxide was started, close the outlet valve on the cylinder, open the stopcocks on the two sampling parts, and remove the cover, and attached 14 inch strips. Immediately cover the exposure chamber with the plain glass cover, close the sampling ports, and continue to draw air through the apparatus for at least an hour to sweep the dinitrogen tetroxide from the mixing and exposure chambers, and attached tubes. Cut the 14 inch strips into 7-inch specimens. Condition the exposed specimens, and the corresponding control specimens by exposure for 20 hours to air at 21°C. and relative humidity at 65 percent. Determine the breaking strength of the exposed and control specimens as specified in Specification CCC-T-191, Method 5100. Calculate the strength loss for each of the exposed specimens from the following formula:

$$\text{Percent breaking strength loss} = 100 - \frac{(E \times 100)}{U}$$

where:

E = breaking strength of exposed specimen.

U = breaking strength of control specimen that was adjacent to the exposed specimen in the original 28-inch strip of fabric.

Repeat the above procedure until all specimens of the fabric have been tested.

4.3.4 *Slippage value—Code No. 12001.*

4.3.4.1 *Apparatus.* For the determination of slippage value, a pendulum type testing machine shall be used as specified in Specification CCC-T-191, Method 5100.

4.3.4.2 *Preparation of specimens.* Specimens, 4 inches wide and 16.0 to 16.5 inches long, shall be taken for test. Two sets of five specimens each are required, one set for warp slippage value, having the short dimension parallel to the warp yarns, and the other set for the filling slippage value having the short dimension parallel to the filling yarns. No two specimens for warp slippage shall contain the same warp yarns and no two specimens for filling slippage shall contain the same filling yarns. Unless otherwise specified, no specimen shall be taken nearer the salvage than one-tenth the width of the material. The specimens shall be prepared by cutting strips of cloth of the number in the directions specified above, approximately 4.5 inches wide and 17 inches long. Any cut or broken threads shall be then raveled from the strip of cloth until only threads remain in the lengthwise direction of the cloth which are approximately the same length as the cloth strip and only threads remain in the short direction which are approximately as long as the width of the cloth strip. Sufficient yarns shall be raveled from both the long and short dimensions of the strip of cloth until the specimen is 4 inches wide by 16 to 16.5 inches long. Any other convenient method of preparing the specimen may be used provided it does not affect the physical properties of the cloth. A test specimen shall be folded back on itself so that the distance from the fold to one edge, parallel to the short dimension, measures approximately 5 inches and to the opposite edge at least 10 inches. The fold must be parallel to the crosswise yarns in the specimen. At 0.38 inch from the fold, a seam shall be sewn parallel to the crosswise yarns of the specimen. The seam shall be of

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the type designated as 301-SSa-1 by Specification DDD-S-751. (The seam SSa-1 shall be made with 1 row of stitch, 12 stitches per inch, Type 301.). The seam shall be made with a suitable needle and a thread of sufficient strength so that it will not break when the seam is tested for elongation as directed below. The specimens shall be conditioned for at least 24 hours at 21° ± 2° C. and 65 ± 2 percent relative humidity.

4.3.4.3 *Procedure*. The distance between the jaws of the test machine shall be adjusted to 3 inches. A 6 ounce clamp shall at attached to the 10 inch length of the specimen, parallel to, end at a distance of 3 inches from the seam described in 4.3.4.2. The clamp can be assembled from a metallic photographic film clip approximately 5 inches long, a brass or steel rod 6 inches long and 3/16 inch in diameter or brass or steel weights, equal in weight, attached to the 6 inch rod to bring the total weight of the assembly to 6 ounces. The test specimen shall be placed with the 10 inch portion uppermost, symmetrically in the upper jaws of the test machine so that the long direction of the specimen is parallel to the direction of the application of the load. The specimen shall be clamped in the upper jaws so that the 6 ounce clamp described above is approximately 1½ inch below the lower jaws of the test machine. The test machine shall be reset so that a zero load is shown on the graphic record. The 6 ounce clamp shall be checked to be certain that it is supported only by the test specimen and that a uniform tension of 6 ounces is applied to the specimen. The lower jaws of the test machine shall be fastened to specimen, care being taken that the same yarns gripped in the lower jaws are gripped in the upper jaws. The apparatus shall be arranged to obtain a tension-stretch curve up to the breaking strength of the specimen, and stretch being indicated in inches by the vertical component of the curve and the tension being indicated in pounds by the horizontal component. The machine shall be started and allowed to apply tension until the specimen breaks. This graphic record shall be designated as the fabric-strength curve. The same specimen shall be inserted in the jaws of the machine so that the seam is in a position parallel to the jaws and midway between the upper and lower jaws. The specimen shall be clamped in the jaws, with tension being applied by means of the 6 ounce clamp, as directed above, before tightening the lower jaws. The apparatus shall be arranged so that a graphic record of this test will be made on the same chart and will start at the same point as the fabric-stretch curve. The machine shall be started and allowed to apply tension to the specimen until either the cloth breaks or the seam fails. This graphic record shall be designated as the seam-stretch curve. The tension shall be noted at which the vertical distance between the seam-stretch curve and the fabric-stretch curve is equal to the sum of:

a. the vertical distance between these curves when the tension is 1 pound, and
b. a vertical distance on the chart equivalent to 0.25 inch on the cloth.

The slippage shall be calculated as follows:

$$\text{Slippage value} = \frac{100\,A}{25}$$

where:

A = tension at which the difference between the seam-stretch curve and the fabric-stretch curve is 0.25 inch plus the difference between these curves when the tension is 1 pound. The slippage value in the warp direction shall be considered as the average of five results obtained for the warp slippage specimens and the slippage value in the filling direction as the average of five results obtained for the filling slippage specimens.

4.3.5 *Stretch—Code No. 13001*. The percent strength shall be determined concurrently with the slippage value and shall be calculated as shown below from measurements of the fabric-stretch curve of the graphic record. The percent of stretch of the warp yarns is obtained when the slippage value of the filling yarns is determined. The percent of stretch of the filling yarns is obtained from the fabric-stretch curve when slippage value of the warp yarn is determined. The percent of stretch in each direction shall be considered as the average of

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five values determined in that direction and shall be calculated as follows:

$$\text{Percent of stretch} = \frac{100\ (A - B)}{3.00 \pm B}$$

where:

A = elongation in inches of the portion of the specimen tested when the tension is 20 pounds as read on the fabric-stretch curve.

B = elongation in inches of the portion of the specimen tested when the tension is 1 pound as read on the fabric stretch curve. The elongation of the specimen is represented by the vertical component of the fabric-stretch curve at the required tension.

4.3.6 *Ash content—Code No. 14001.* Parts from each of the pieces comprising the sample shall be cut, so as to obtain a total of approximately 2 grams (gm.) transferred to a tared porcelain crucible, and moistened with concentrated nitric acid. The crucible shall be heated on a steam bath for 1 hour and then heated carefully over a flame so as to avoid any mechanical loss. When the contents of the crucible have been charred, it shall be heated to dull redness until all carbonceous matter has been burned off. The crucible shall be cooled in a dessicator, and weighed. The ash content shall be calculated as follows:

$$\text{Percent ash} = \frac{100\ (A - B)}{W}$$

where:

A = weight of crucible and ignited residue.

B = weight of crucible empty.

W = weight of sample.

4.3.7 *Color.* Code No. 15001.

4.3.7.1 *Colors 1 and 2.* The dyed or tinted fabric shall be scoured or stripped, to remove color only, in accordance with a recommended commercial procedure for the particular color. The dye or tint shall be considered fugitive if all the color is removed from the fabric (see 6.5).

4.3.7.2 *Class 3.* In all instances the cloth shall be dyed scarlet. The scarlet dye shall produce a color closely approximating the color No. 1105 of Fed-STD-595. The dyestuffs used shall be either of the following or a blend of the two:

Sodium salt of ditolyl-disazo-bis-naphthylamine-4 sulfonic acid $C_{34}H_{18}N_6S_2Na_2$

Note. The color index number is 448—Society of Dyers and Colorists.

Sodium salt of 4 sulfo-naphthalene-azo-B-napthol $C_{20}H_{13}N_2O4SNa$

Note. The color index number is 176. Society of Dyers and Colorists.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. (See 6.2.)

5.1.1 *Level A.* The cartridge cloth shall be furnished in bolts wrapped in waxed paper in accordance with Specification UU-P-270, Type II, Class 3.

5.2 Packing. (See 6.2.)

5.2.1 *Level A.* The packages containing cartridge cloth shall be packed in cleated plywood boxes, nailed wood boxes, or wirebound wood boxes, conforming to the requirements of Specification PPP-B-601, PPP-B-621 or PPP-B-585, respectively. Shipping containers shall be lined with a sealed waterproof case liner complying with specification MIL-L-10547, or equivalent. Containers shall be strapped in accordance with the strapping requirements of the appendix of the applicable box specification. The gross weight shall not exceed 200 pounds.

5.2.2 *Level C.* The cartridge cloth shall be packed for shipment in conformance with consolidated freight classification rules and container specification for rail shipments, or with National Motor Freight Classification rules and container specifications for truck shipment.

5.3 Marking. Interior packages, and exterior containers, shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. The cartridge cloths are intended for use in manufacture of cartridge bags for artillery propelling charges.

6.2 Ordering date. Procurement documents should specify:

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(a) Title, number and date of this specification.
(b) Class of cloth required.
(c) Width of material required. (see 3.3).
(d) Color of material required. (see 3.12).
(e) Levels of protection. (see 5.1).

6.3 **Finish.** One finish that has been found satisfactory is the Permel Plus finish — American Cyanamid Co. This finish deposits on the desized cloth, dry basis, 12 to 14 percent solids urea formaldehyde resin (Aerotex cream 450) and 6–8 percent solids melamine resin base (Permel B).

6.3.1. *Processing.* The processing conditions that give the above finish are as follows:

Impregnating Bath Composition	*Percent*
Aerotex Cream 450	20
Permel Resin B	15
Aerotex Resin 159	6
Aerotex Buffer DCY	0.54
Aerotex Accelerator	1.00

Impregnation:
Immersion: 2 dips 2 nips
Pressure: 2.9 tons
Expression: 100 percent

Dry and Cure:
Dry at 225°F.
Cure at 300°F. for 3 to 4 minutes.

6.4 **Breaking strength loss.** Crowding of the exposure chamber has been found to give poor reproducibility and high values for dinitrogen tetroxide resistance. It is recommended, therefore, that no more than 6 different fabrics be exposed together and that one of these be a cotton fabric whose dinitrogen tetroxide resistance has been determined previously. Failure of the value obtained for the dinitrogen tetroxide resistance of this fabric to agree with the values previously obtained would indicate that a change has been made in the conditions of the test, or that one of the fabrics present is in some way interfering with the test. As a check on the latter point, the fabrics should be exposed separately (see 4.3.3.2).

6.5 **Color.** The object of the fugitive color is to preclude the masking of results under the stain test (see 3.12).

6.6 **Equipment.** Equipment required to conduct this test is available at Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-ND 4. This test shall be conducted at Government expense or at no expense to the contractor.

6.7 **Inspection code numbers.** The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.8 The resin finish cloth, when stained in accordance with Paragraph 4.3.2 shall give a uniform purple color.

Custodian:
ARMY—MU

Preparing Activity:
ARMY—MU
Project No. 8305–A052Y

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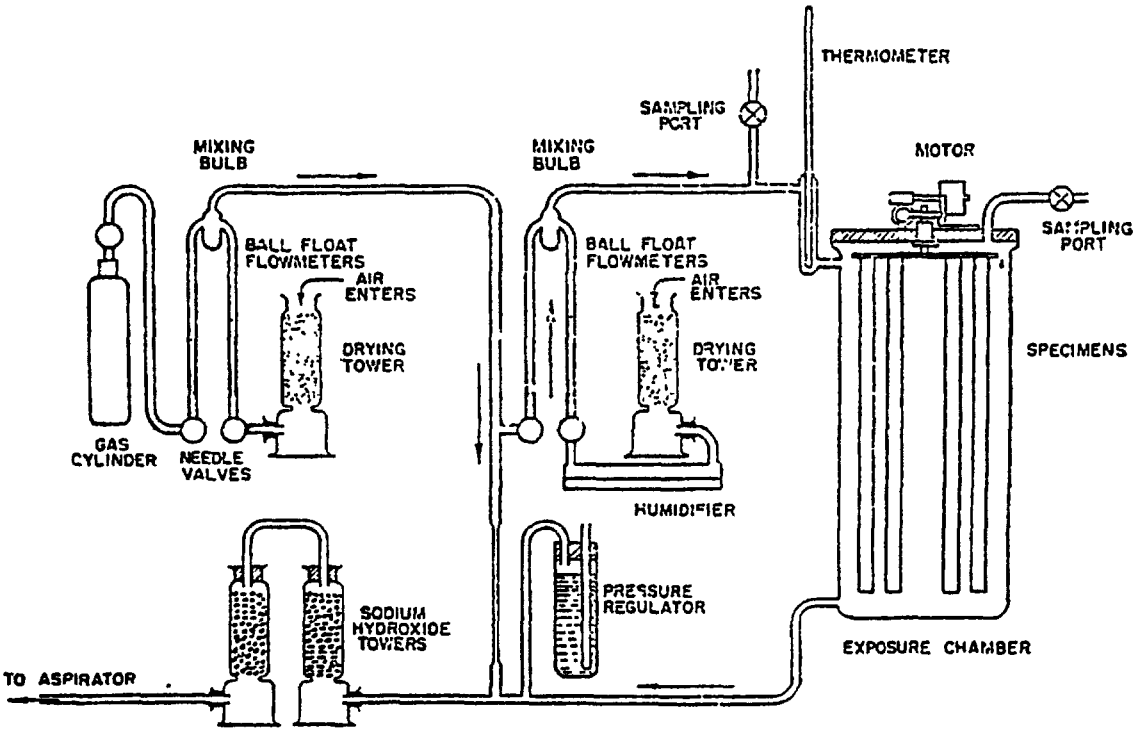


FIGURE 1. *Apparatus for exposing fabric specimens to dinitrogen tetroxide.*

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5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

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